Bridging functional and network robustness

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Summary

The idea

The role of a species in a community can be defined in three ways: where the species is in the functional space, where the species is in the ecological network, and what is the impact of the species loss. These are three lenses with which we can look at a species. They are usually used separately. Yet, they all are an attempt to capture the very same reality. So we argue that to improve our view on a species role in a community it is best to pile up these lenses than using them separately. They are complementary facets and understanding the relationships between these different view enable a more complete view of a role played by a species within a community.

The question

To what degree do a species position in functional trait space, its structural position in the interaction network, and its contribution to community robustness converge or diverge? What does that tell us about community robustness?

The axes

1. From static correlation to mechanistic understanding

Coux et al. (2016) established that functional originality correlates with network specialisation in pollinators. But the why remains underexplored. Traits govern which interactions are possible, such as morphological matching between flower and pollinators. So the trait–network relationship is not merely correlative but mechanistic. A promising direction is to move beyond summary axes (PCoA) and ask which specific traits drive which network metrics. Disentangling these trait-by-metric relationships would give the correlation a mechanistic foundation.

2. Connecting functional position to extinction dynamics

A key question whether functional originality of the lost species determines the magnitude of that loss. If a species is functionally original and topologically central, does its removal trigger a disproportionate cascade? Or do these two dimensions predict different things?

3. Generality across systems

Coux et al. (2016) worked exclusively with plant-pollinator networks, an extension is to test whether the same framework holds in other interaction types — food webs, seed dispersal networks — where the trait–interaction link operates differently. If the relationship between functional position and network role is general, it points to a universal principle of community organisation. If it is system-specific, that contrast itself becomes informative about how different ecological architectures buffer or amplify the consequences of species loss.

Framework

Conceptual overview

We decompose the question of species robustness into three interconnected facets: functional trait space position, structural network role, and extinction dynamics. Our analysis links these facets at the species level across empirical plant–pollinator networks (Coux et al. 2016).

Literature landscape

Data and species descriptors

From each network we extract three types of species-level descriptors.

Trait-space descriptors (following Coux et al. (2016); Violle et al. (2017)) summarise a species' position in its own functional trait space

- *functional originality*, that is, the distance of species i to the community centroid in PCoA trait space

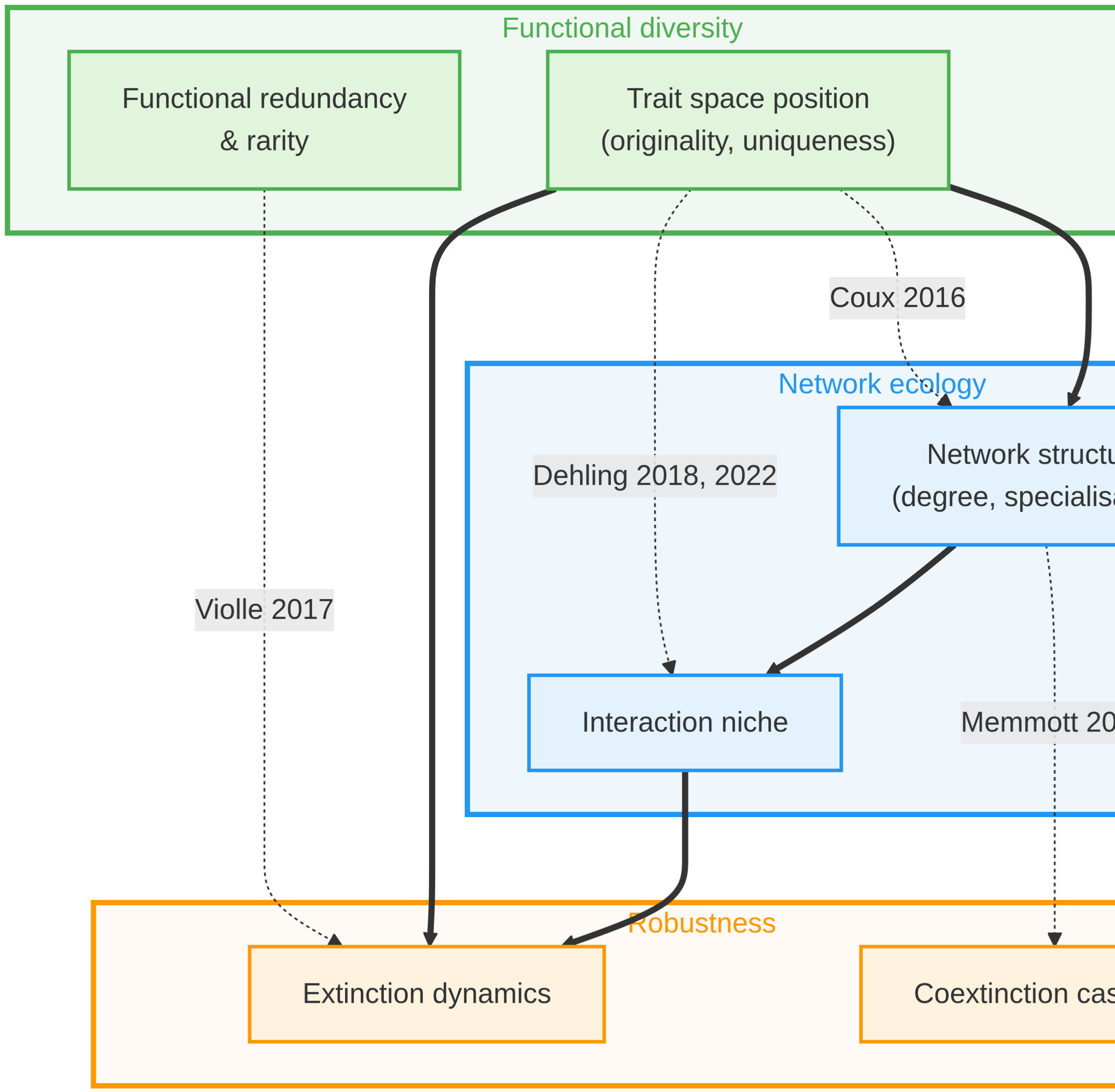


Figure 1: Literature landscape showing three research fields and their partial connections. Network ecology and functional diversity have been partially bridged through studies linking trait space position to network structure. Functional diversity and robustness connect through functional redundancy concepts. Network ecology and robustness link via structural importance measures. These three perspectives remain to be integrated all at the same time.

- *functional uniqueness*, that is, distance to the nearest neighbour in trait space

Network metrics characterise structural embeddedness in the interaction network

- *degree* k_i
- *Blüthgen's* d'_i (Blüthgen et al. 2006), that is, the deviation of interaction distribution from random given partner abundances

Interaction niche metrics (following Dehling and Stouffer (2018); Dehling et al. (2022)) characterise a species through the traits of its interaction partners

- *interaction niche position*, that is, the centroid of species i in partner trait space
- *interaction niche size*, that is, the hypervolume spanned by interaction partners in trait space
- *interaction niche originality*, that is, distance of species i's interaction centroid from the community mean interaction centroid

Extinction simulation

We simulate pollinator extinctions using both a topological coextinction model (Memmott et al. 2004) and a stochastic coextinction model (Vieira and Almeida-Neto 2015), implemented via the `bipartite` package (Dormann et al. 2008) in R.

For each species i and each network, we compute three per-species extinction impact metrics

$$\Delta R_i = R_{\text{full}} - R_{-i}$$

where R_{-i} is network robustness when species i is removed first,

$$\text{cascade}_i = \mathbb{E}[\text{secondary extinctions triggered by removal of } i]$$

averaged over 1000 stochastic realisations, and

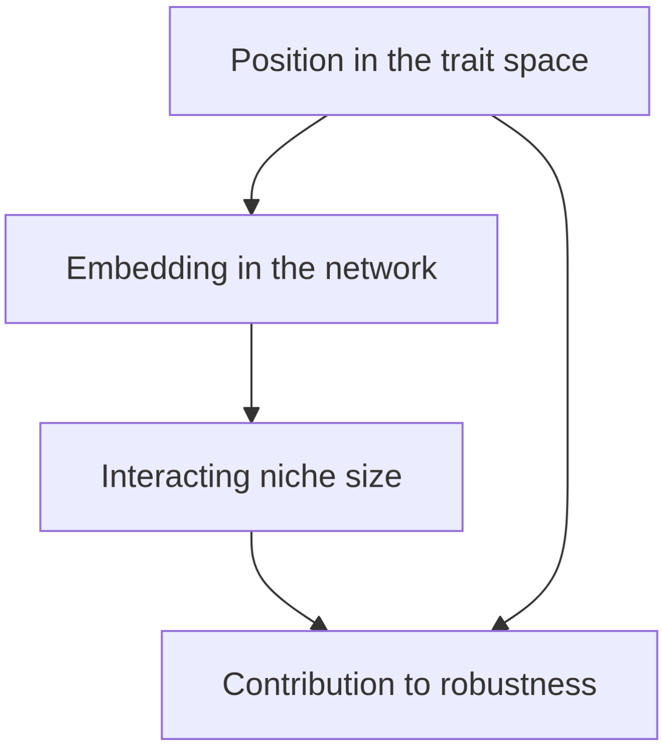
$$\Delta \text{FD}_i = \text{FD}_{\text{full}} - \text{FD}_{-i}$$

the loss of plant functional diversity following removal of i and its downstream secondary extinctions.

Simulations are run under three extinction sequences – random, least-to-most-abundant, specialist-to-generalist – to assess robustness of conclusions to sequence choice.

Causal structure and statistical model

The predictors above are not causally independent. A species' own traits constrain which plants it can interact with (trait matching), which determines its network degree and specialisation, which in turn shapes its interaction niche size and modularity role. We therefore adopt a **structural equation modelling** which accounts for the causal structure.



The hypothesised causal graph contains four paths

1. originality $\rightarrow$ degree [trait matching]
2. degree $\rightarrow$ interaction niche size [bridging Eltonian niche and functional diversity]
3. interaction niche size $\rightarrow$ ΔR
4. originality $\rightarrow$ ΔR [functional redundancy]

Other ideas

Invasive species

Link with invasive species. Invasive species may have a high trait uniqueness which make them able of taking a place that is not fulfilled in the pristine community. That is, can we link the position of the invasive species in the trait space to its probability to invade? In the same vein, can we also relate the position in the trait space of the invasive species to its impact on the community?

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